

CSL351: Computer Networks

Assignment - 1 (Basic Network Tools)

ID No: 12340920

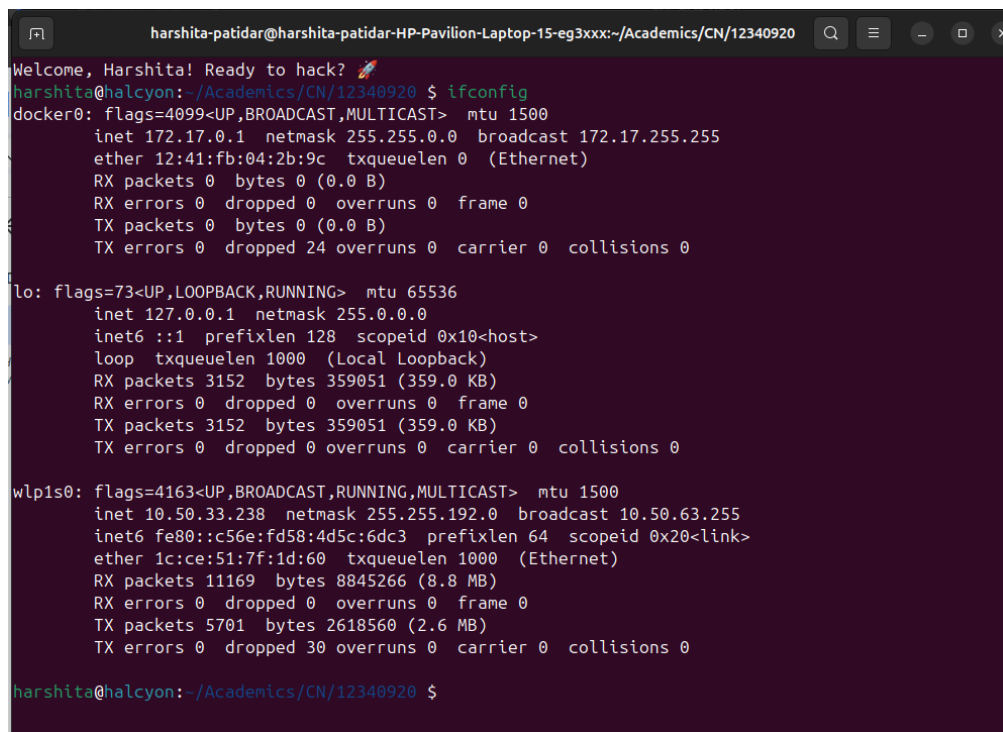
Name: Harshita Patidar

1 Question 1: IFCONFIG Command

a. Command and Output Description

Command:

```
ifconfig
```



```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx:~/Academics/CN/12340920
Welcome, Harshita! Ready to hack? 🚀
harshita@halcyon: ~/Academics/CN/12340920 $ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 12:41:fb:04:2b:9c txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 24 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 3152 bytes 359051 (359.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3152 bytes 359051 (359.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.50.33.238 netmask 255.255.192.0 broadcast 10.50.63.255
    inet6 fe80::c56e:fd58:4d5c:6dc3 prefixlen 64 scopeid 0x20<link>
    ether 1c:ce:51:7f:1d:60 txqueuelen 1000 (Ethernet)
    RX packets 11169 bytes 8845266 (8.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5701 bytes 2618560 (2.6 MB)
    TX errors 0 dropped 30 overruns 0 carrier 0 collisions 0

harshita@halcyon: ~/Academics/CN/12340920 $
```

Figure 1: Output of command: ifconfig

Explanation of Output: The `ifconfig` command displays the status and configuration details of all active network interfaces. On my system, the following interfaces were observed:

- **docker0:** No packets received or transmitted which indicates that no active Docker container traffic at the time when I ran the command.

- **lo (Loopback Interface):** active RX and TX packets are found, confirming that local processes are communicating through the loopback interface.
- **wlp1s0:** Shows RX and TX packet counts, indicating active network usage.

Important Attributes:

- **inet/inet6:** IPv4 and IPv6 addresses assigned to the interface, used to identify our device on a network so data can be sent and received.
- **Ether:** MAC (hardware) address of a network interface, used to identify it at the physical (link) layer. Docker shows ether because every interface must have a MAC address, and IPv6 is disabled by default in Docker.
- **netmask:** 32-bit number that divides an IP address into two parts: the network ID (identifying the specific network) and the host ID (identifying the device on that network)
- **RX/TX packets:** RX packets are data packets our device receives, and TX packets are data packets our device sends through a network interface.
- **MTU:** (Maximum Transmission Unit) is the largest size of a data packet that a network interface can send or receive at one time without breaking it into smaller pieces.
- **UP/RUNNING:** Indicates the current operational state.
- **Broadcast:** special network address used to send a packet to all devices on the same local network at the same time.
- **Errors:** Errors means packets that were received or sent incorrectly (damaged or invalid)
- **Dropped:** means packets that were discarded because the system or network couldn't handle them (e.g., buffer full or congestion).

b. ifconfig Options

1. **-a:** Displays all interfaces, including those that are currently down.
2. **-s:** Displays a short list of interfaces.
3. **-v:** more detailed output for some error conditions
4. **up:** activates the interface (it can send/receive data)
5. **down:** deactivates the interface (it cannot send/receive data)

```

harshita@harshita-patidar-HP-Pavilion-Laptop-...
harshita@halcyon:~/Academics/CN/12340920 $ ifconfig -a
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 12:41:fb:04:2b:9c txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 868 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 83864 bytes 12915364 (12.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 83864 bytes 12915364 (12.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.72.48 netmask 255.255.240.0 broadcast 10.10.79.255
    inet6 fe80::b71b:10fa:c7f9:b07e prefixlen 64 scopeid 0x20<link>
    ether 1c:ce:51:7f:1d:60 txqueuelen 1000 (Ethernet)
    RX packets 2128238 bytes 2340176492 (2.3 GB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 885461 bytes 297298395 (297.2 MB)
    TX errors 0 dropped 340 overruns 0 carrier 0 collisions 0

harshita@halcyon:~/Academics/CN/12340920 $ ifconfig -s
Iface MTU RX-OK RX-ERR RX-DRP RX-OVR TX-OK TX-ERR TX-DRP TX-OVR Flg
docker0 1500 0 0 0 0 0 0 0 869 0 BMU
lo 65536 83878 0 0 0 83878 0 0 0 0 LRU
wlp1s0 1500 2128402 0 0 0 885606 0 0 340 0 BMRU

harshita@halcyon:~/Academics/CN/12340920 $ ifconfig -v
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 12:41:fb:04:2b:9c txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 869 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 83878 bytes 12918315 (12.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 83878 bytes 12918315 (12.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.10.72.48 netmask 255.255.240.0 broadcast 10.10.79.255
    inet6 fe80::b71b:10fa:c7f9:b07e prefixlen 64 scopeid 0x20<link>
    ether 1c:ce:51:7f:1d:60 txqueuelen 1000 (Ethernet)
    RX packets 2128422 bytes 2340242209 (2.3 GB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 885621 bytes 297348534 (297.3 MB)
    TX errors 0 dropped 340 overruns 0 carrier 0 collisions 0

harshita@halcyon:~/Academics/CN/12340920 $ |

```

2 Question 2: NETSTAT Command

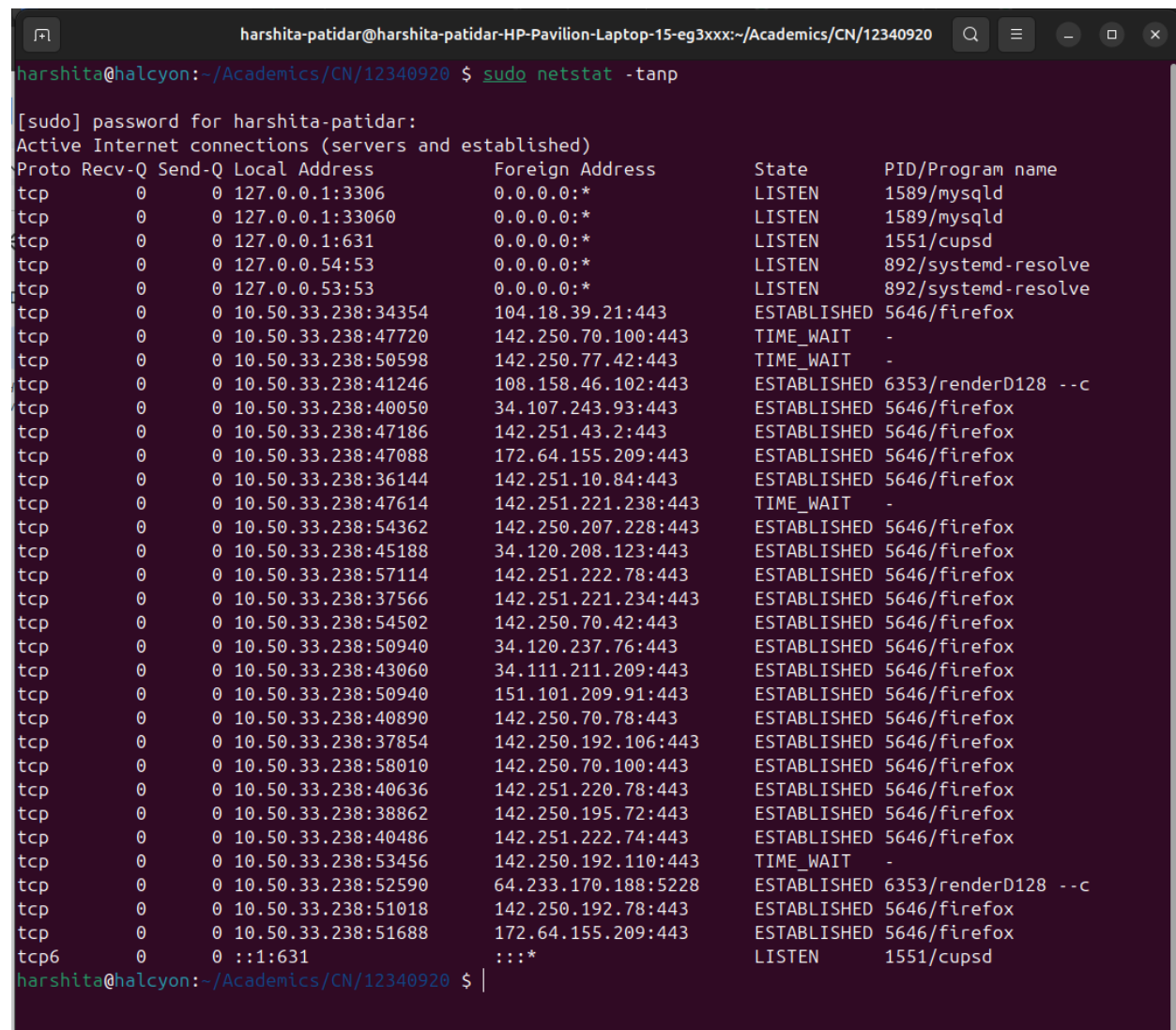
a. Use of Netstat

The `netstat` command is used to know network connections, listening services, routing information, and network statistics of your system.

b. Active TCP Ports and Browser Identification

Command to find all active TCP ports:

```
sudo netstat -tanp
```



```
harshita@halcyon:~/Academics/CN/12340920 $ sudo netstat -tanp
[sudo] password for harshita-patidar:
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:3306          0.0.0.0:*               LISTEN      1589/mysqld
tcp        0      0 127.0.0.1:33060        0.0.0.0:*               LISTEN      1589/mysqld
tcp        0      0 127.0.0.1:631          0.0.0.0:*               LISTEN      1551/cupsd
tcp        0      0 127.0.0.54:53          0.0.0.0:*               LISTEN      892/systemd-resolve
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN      892/systemd-resolve
tcp        0      0 10.50.33.238:34354     104.18.39.21:443        ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:47720     142.250.70.100:443      TIME_WAIT   -
tcp        0      0 10.50.33.238:50598     142.250.77.42:443      TIME_WAIT   -
tcp        0      0 10.50.33.238:41246     108.158.46.102:443      ESTABLISHED 6353/renderD128 --c
tcp        0      0 10.50.33.238:40050     34.107.243.93:443      ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:47186     142.251.43.2:443       ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:47088     172.64.155.209:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:36144     142.251.10.84:443      ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:47614     142.251.221.238:443    TIME_WAIT   -
tcp        0      0 10.50.33.238:54362     142.250.207.228:443    ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:45188     34.120.208.123:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:57114     142.251.222.78:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:37566     142.251.221.234:443    ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:54502     142.250.70.42:443      ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:50940     34.120.237.76:443      ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:43060     34.111.211.209:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:50940     151.101.209.91:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:40890     142.250.70.78:443      ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:37854     142.250.192.106:443    ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:58010     142.250.70.100:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:40636     142.251.220.78:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:38862     142.250.195.72:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:40486     142.251.222.74:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:53456     142.250.192.110:443    TIME_WAIT   -
tcp        0      0 10.50.33.238:52590     64.233.170.188:5228    ESTABLISHED 6353/renderD128 --c
tcp        0      0 10.50.33.238:51018     142.250.192.78:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:51688     172.64.155.209:443     ESTABLISHED 5646/firefox
tcp6       0      0 :::1:631               :::*                   LISTEN      1551/cupsd
harshita@halcyon:~/Academics/CN/12340920 $
```

Identify Web Browser PID/Ports:

```
sudo netstat -tanp | grep -i firefox # or chrome
```

```
harshita@halcyon:~/Academics/CN/12340920 $ sudo netstat -tanp | grep -E firefox
tcp        0      0 10.50.33.238:34354    104.18.39.21:443      ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:40050    34.107.243.93:443     ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:34280    142.250.195.14:443    ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:47088    172.64.155.209:443    ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:45660    172.64.155.209:443    ESTABLISHED 5646/firefox
tcp        0      0 10.50.33.238:51140    142.250.195.14:443    ESTABLISHED 5646/firefox
harshita@halcyon:~/Academics/CN/12340920 $
```

Tab Identification: Tab identification is not possible using netstat alone, as modern browsers employ a multi-process architecture where a single process may serve multiple tabs, and a single tab may span multiple processes.

Standard Service Check:

Service	Port	Found?	Identification
DNS	53	Yes	systemd-resolve
HTTP	80	No	No entry found
DHCP	67/68	No	No entry found
SMTP	25	No	No entry found
FTP	21	No	No entry found

c. UDP Statistics

Command:

```
netstat -su
```

Observation: The system shows normal UDP and ICMP traffic with a small number of dropped packets, which is expected on an active Wi-Fi network.

```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15...
harshita@halcyon:~/Academics/CN/12340920 $ netstat -su
IcmpMsg:
  InType3: 168
  InType8: 1
  OutType0: 1
  OutType3: 188
Udp:
  40056 packets received
  143 packets to unknown port received
  64 packet receive errors
  24727 packets sent
  64 receive buffer errors
  3 send buffer errors
UdpLite:
IpExt:
  InNoRoutes: 70
  InMcastPkts: 164
  OutMcastPkts: 227
  InOctets: 94887328
  OutOctets: 32245056
  InMcastOctets: 19331
  OutMcastOctets: 28951
  InNoECTPkts: 97557
  InECT1Pkts: 10861
  InECT0Pkts: 493
MPTcpExt:
harshita@halcyon:~/Academics/CN/12340920 $
```

3 Question 3: PING Command

a. Use of Ping

Ping uses **ICMP Echo Request/Reply** messages to verify connectivity. The primary functions are:

- **Connectivity Testing:** Verifying if a remote host is reachable.
- **Latency Measurement:** Calculating the Round-Trip Time (RTT) to evaluate network speed.

- **Packet Loss Analysis:** Identifying network instability or congestion by tracking failed replies.
- **DNS Verification:** Confirming hostname-to-IP resolution.

b. Experiments

Time: 10:48 PM

```

Thu Jan 8 10:48:40 PM
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx:~/Academics/CN/12340920
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 google.com
PING google.com (142.250.207.174) 56(84) bytes of data.
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=2 ttl=113 time=186 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=3 ttl=113 time=108 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=4 ttl=113 time=140 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=5 ttl=113 time=152 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=7 ttl=113 time=146 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=8 ttl=113 time=169 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=9 ttl=113 time=192 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=10 ttl=113 time=112 ms

--- google.com ping statistics ---
10 packets transmitted, 8 received, 20% packet loss, time 9088ms
rtt min/avg/max/mdev = 107.611/150.466/191.814/29.108 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 iitbhilai.ac.in
PING iitbhilai.ac.in (192.168.10.115) 56(84) bytes of data.
64 bytes from 192.168.10.115: icmp_seq=1 ttl=62 time=101 ms
64 bytes from 192.168.10.115: icmp_seq=2 ttl=62 time=123 ms
64 bytes from 192.168.10.115: icmp_seq=3 ttl=62 time=146 ms
64 bytes from 192.168.10.115: icmp_seq=4 ttl=62 time=169 ms
64 bytes from 192.168.10.115: icmp_seq=5 ttl=62 time=88.3 ms
64 bytes from 192.168.10.115: icmp_seq=7 ttl=62 time=92.3 ms
64 bytes from 192.168.10.115: icmp_seq=8 ttl=62 time=115 ms
64 bytes from 192.168.10.115: icmp_seq=9 ttl=62 time=136 ms
64 bytes from 192.168.10.115: icmp_seq=10 ttl=62 time=159 ms

--- iitbhilai.ac.in ping statistics ---
10 packets transmitted, 9 received, 10% packet loss, time 9056ms
rtt min/avg/max/mdev = 88.334/125.518/168.935/27.460 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 cloudflare.com
PING cloudflare.com (104.16.132.229) 56(84) bytes of data.
64 bytes from 104.16.132.229: icmp_seq=1 ttl=51 time=190 ms
64 bytes from 104.16.132.229: icmp_seq=2 ttl=54 time=111 ms
64 bytes from 104.16.132.229: icmp_seq=3 ttl=54 time=133 ms
64 bytes from 104.16.132.229: icmp_seq=4 ttl=54 time=156 ms
64 bytes from 104.16.132.229: icmp_seq=5 ttl=54 time=177 ms
64 bytes from 104.16.132.229: icmp_seq=7 ttl=54 time=166 ms
64 bytes from 104.16.132.229: icmp_seq=8 ttl=54 time=189 ms
64 bytes from 104.16.132.229: icmp_seq=9 ttl=54 time=210 ms
64 bytes from 104.16.132.229: icmp_seq=10 ttl=54 time=130 ms

--- cloudflare.com ping statistics ---
10 packets transmitted, 9 received, 10% packet loss, time 9071ms
rtt min/avg/max/mdev = 110.643/162.231/209.694/30.884 ms
harshita@halcyon:~/Academics/CN/12340920 $ |

```

Time: 10:21 AM

```
Fri Jan 9 10:21:05 AM
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx:...
Welcome, Harshita! Ready to hack? 🚀
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 iitbhillai.ac.in
PING iitbhillai.ac.in (192.168.10.115) 56(84) bytes of data.
64 bytes from 192.168.10.115: icmp_seq=1 ttl=62 time=12.5 ms
64 bytes from 192.168.10.115: icmp_seq=2 ttl=62 time=2.33 ms
64 bytes from 192.168.10.115: icmp_seq=3 ttl=62 time=8.11 ms
64 bytes from 192.168.10.115: icmp_seq=4 ttl=62 time=2.41 ms
64 bytes from 192.168.10.115: icmp_seq=5 ttl=62 time=6.89 ms
64 bytes from 192.168.10.115: icmp_seq=6 ttl=62 time=125 ms
64 bytes from 192.168.10.115: icmp_seq=7 ttl=62 time=7.41 ms
64 bytes from 192.168.10.115: icmp_seq=8 ttl=62 time=67.4 ms
64 bytes from 192.168.10.115: icmp_seq=9 ttl=62 time=12.2 ms
64 bytes from 192.168.10.115: icmp_seq=10 ttl=62 time=5.64 ms

--- iitbhillai.ac.in ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9015ms
rtt min/avg/max/mdev = 2.325/24.987/125.046/38.009 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 google.com
PING google.com (142.250.207.174) 56(84) bytes of data.
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=1 ttl=113 time=99.5 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=2 ttl=113 time=22.0 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=3 ttl=113 time=24.7 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=4 ttl=113 time=61.4 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=5 ttl=113 time=78.2 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=6 ttl=113 time=106 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=7 ttl=113 time=27.1 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=8 ttl=113 time=45.4 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=9 ttl=113 time=68.0 ms
64 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=10 ttl=113 time=196 ms

--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9016ms
rtt min/avg/max/mdev = 21.965/72.814/195.554/49.835 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 cloudflare.com
PING cloudflare.com (104.16.132.229) 56(84) bytes of data.
64 bytes from 104.16.132.229: icmp_seq=1 ttl=54 time=108 ms
64 bytes from 104.16.132.229: icmp_seq=2 ttl=54 time=237 ms
64 bytes from 104.16.132.229: icmp_seq=3 ttl=54 time=28.1 ms
64 bytes from 104.16.132.229: icmp_seq=4 ttl=54 time=74.6 ms
64 bytes from 104.16.132.229: icmp_seq=5 ttl=54 time=44.9 ms
64 bytes from 104.16.132.229: icmp_seq=6 ttl=54 time=123 ms
64 bytes from 104.16.132.229: icmp_seq=7 ttl=54 time=146 ms
64 bytes from 104.16.132.229: icmp_seq=8 ttl=54 time=65.1 ms
64 bytes from 104.16.132.229: icmp_seq=9 ttl=54 time=65.6 ms
64 bytes from 104.16.132.229: icmp_seq=10 ttl=54 time=109 ms

--- cloudflare.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
rtt min/avg/max/mdev = 28.072/100.245/237.029/57.323 ms
harshita@halcyon:~/Academics/CN/12340920 $
```

Time: 2:59 PM


```

Fri Jan 9 2:59:13 PM
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx:...
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 iitbhilai.ac.in
PING iitbhilai.ac.in (192.168.10.115) 56(84) bytes of data.
64 bytes from 192.168.10.115: icmp_seq=1 ttl=62 time=3.38 ms
64 bytes from 192.168.10.115: icmp_seq=2 ttl=62 time=15.5 ms
64 bytes from 192.168.10.115: icmp_seq=3 ttl=62 time=4.16 ms
64 bytes from 192.168.10.115: icmp_seq=4 ttl=62 time=8.29 ms
64 bytes from 192.168.10.115: icmp_seq=5 ttl=62 time=2.92 ms
64 bytes from 192.168.10.115: icmp_seq=6 ttl=62 time=5.11 ms
64 bytes from 192.168.10.115: icmp_seq=7 ttl=62 time=2.78 ms
64 bytes from 192.168.10.115: icmp_seq=8 ttl=62 time=8.63 ms
64 bytes from 192.168.10.115: icmp_seq=9 ttl=62 time=2.95 ms
64 bytes from 192.168.10.115: icmp_seq=10 ttl=62 time=1.70 ms

--- iitbhilai.ac.in ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9015ms
rtt min/avg/max/mdev = 1.702/5.543/15.510/3.988 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 google.com
PING google.com (142.251.43.14) 56(84) bytes of data.
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=1 ttl=113 time=104 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=2 ttl=113 time=126 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=3 ttl=113 time=45.3 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=4 ttl=113 time=67.4 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=5 ttl=113 time=89.6 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=6 ttl=113 time=112 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=7 ttl=113 time=32.1 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=8 ttl=113 time=54.8 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=9 ttl=113 time=77.6 ms
64 bytes from tsao3s08-in-f14.1e100.net (142.251.43.14): icmp_seq=10 ttl=113 time=99.9 ms

--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
rtt min/avg/max/mdev = 32.068/80.949/126.429/29.247 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 cloudflare.com
PING cloudflare.com (104.16.132.229) 56(84) bytes of data.
64 bytes from 104.16.132.229: icmp_seq=1 ttl=51 time=121 ms
64 bytes from 104.16.132.229: icmp_seq=2 ttl=54 time=42.4 ms
64 bytes from 104.16.132.229: icmp_seq=3 ttl=54 time=64.0 ms
64 bytes from 104.16.132.229: icmp_seq=4 ttl=54 time=87.2 ms
64 bytes from 104.16.132.229: icmp_seq=5 ttl=54 time=110 ms
64 bytes from 104.16.132.229: icmp_seq=6 ttl=54 time=132 ms
64 bytes from 104.16.132.229: icmp_seq=7 ttl=54 time=53.6 ms
64 bytes from 104.16.132.229: icmp_seq=8 ttl=54 time=74.1 ms
64 bytes from 104.16.132.229: icmp_seq=9 ttl=54 time=96.7 ms
64 bytes from 104.16.132.229: icmp_seq=10 ttl=54 time=119 ms

--- cloudflare.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
rtt min/avg/max/mdev = 42.396/90.046/132.035/29.262 ms
harshita@halcyon:~/Academics/CN/12340920 $

```

i. Average RTT vs Geographical Distance

Host	Avg RTT-Morning	Avg RTT-Afternoon	Avg RTT-Night
iitbhilai.ac.in	24.98 ms	5.543 ms	125.518 ms
google.com	72.814 ms	80.949 ms	150.466 ms
cloudflare.com	100.245 ms	90.046 ms	162.231 ms

Conclusion: RTT generally increases with geographical distance, but it is not the only factor.

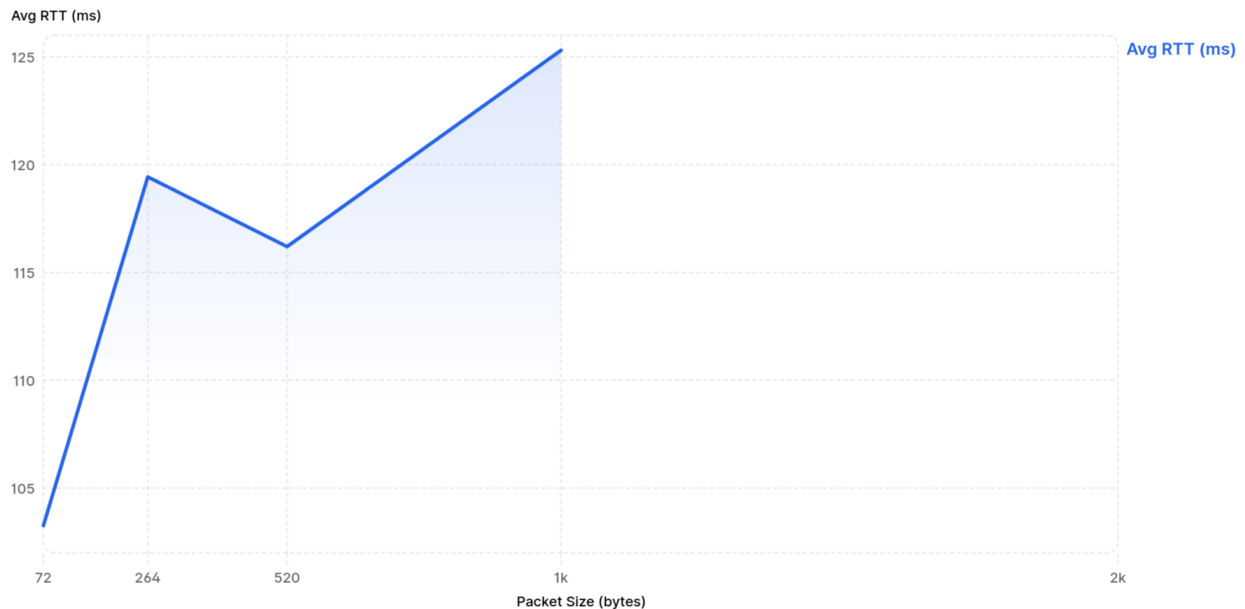
- Since the system is connected to IIT Bhilai's Wi-Fi network, the RTT to iitbhilai.ac.in is very low.
- However, a significant increase in RTT for iitbhilai.ac.in is observed during night time. This may be due to a large no of students and other users actively using the campus network during night time which leads to increased network congestion and queueing delay on internal routers and access points.
- google.com and cloudflare.com are external Internet hosts located far from the source network. Thus, show higher and variable RTT values influenced by distance and network load.

ii. Packet Size vs RTT

Packet Size (Bytes)	Avg RTT (ms)
72	103.259
264	119.433
520	116.198
1024	125.303
2048	NIL (100% Loss)

Average R T T vs packet size.

Analysis



Thu Jan 8 11:21:58 PM

```
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 -s 64 google.com
PING google.com (142.250.207.174) 64(92) bytes of data.
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=1 ttl=113 time=101 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=2 ttl=113 time=123 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=3 ttl=113 time=146 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=4 ttl=113 time=168 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=5 ttl=113 time=88.7 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=6 ttl=113 time=111 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=7 ttl=113 time=134 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=8 ttl=113 time=50.5 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=9 ttl=113 time=76.9 ms
72 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=10 ttl=113 time=32.6 ms

--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 901ms
rtt min/avg/max/ndev = 32.568/103.259/168.498/40.226 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 -s 256 google.com
PING google.com (142.250.207.174) 256(284) bytes of data.
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=1 ttl=113 time=88.5 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=2 ttl=113 time=146 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=3 ttl=113 time=67.2 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=4 ttl=113 time=89.9 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=5 ttl=113 time=112 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=6 ttl=113 time=134 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=7 ttl=113 time=54.2 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=8 ttl=113 time=179 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=9 ttl=113 time=201 ms
264 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=10 ttl=113 time=123 ms

--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 901ms
rtt min/avg/max/ndev = 54.230/119.433/200.984/44.693 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 -s 512 google.com
PING google.com (142.250.207.174) 512(540) bytes of data.
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=1 ttl=113 time=100 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=2 ttl=113 time=123 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=3 ttl=113 time=146 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=4 ttl=113 time=87.2 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=5 ttl=113 time=88.2 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=6 ttl=113 time=111 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=7 ttl=113 time=133 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=8 ttl=113 time=117 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=9 ttl=113 time=141 ms
520 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=10 ttl=113 time=141 ms

--- google.com ping statistics ---
10 packets transmitted, 9 received, 10% packet loss, time 907ms
rtt min/avg/max/ndev = 87.166/116.198/146.234/20.391 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 -s 1024 google.com
PING google.com (142.250.207.174) 1024(1052) bytes of data.
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=1 ttl=113 time=106 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=2 ttl=113 time=102 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=3 ttl=113 time=125 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=4 ttl=113 time=148 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=5 ttl=113 time=68.7 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=6 ttl=113 time=92.7 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=7 ttl=113 time=116 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=8 ttl=113 time=140 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=9 ttl=113 time=164 ms
1032 bytes from pnbomb-bl-in-f14.1e100.net (142.250.207.174): icmp_seq=10 ttl=113 time=191 ms

--- google.com ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 1323ms
rtt min/avg/max/ndev = 68.670/125.303/191.191/34.450 ms
harshita@halcyon:~/Academics/CN/12340920 $ ping -c 10 -s 2048 google.com
PING google.com (142.250.207.174) 2048(2076) bytes of data.

--- google.com ping statistics ---
10 packets transmitted, 0 received, 100% packet loss, time 9214ms
```

Observation: For 2048 bytes, no reply was received. This indicates that the packet size exceeded the allowed MTU on the path

iii. Impacts on RTT

- **Packet Size:** Generally, with increase in packet size, RTT also increases. Large sized packets takes more time to transmit and they may undergo fragmentation at some intermediate routers, which adds processing and queueing delays.
- **Time of Day:** RTT varies with the time of day due to network load. During peak hours (especially at night on campus networks like IIT Bhilai), more users are active, causing congestion, increased queueing delays, and higher RTT. During off-peak hours, RTT is lower and more stable due to reduced traffic.

4 Question 4: TRACEROUTE

a. Use of Traceroute

Traceroute is a tool which is used to track the path that data packets follow from the source to a destination. It shows each intermediate router (hop) along the route and the time taken to reach them. This helps in understanding the network path, identifying delays, packet loss, or points where the connection fails, and is especially useful for troubleshooting network connectivity issues.

b. Incomplete Paths (* * *)

1. Some routers or destination hosts block ICMP/UDP packets used by traceroute for security reasons. 2. Firewalls may drop traceroute packets, resulting in * * * in the output. 3. Routers may rate-limit ICMP responses, so they do not reply to every probe. 4. Network congestion or overloaded routers can cause timeouts.

c. Is it possible to find the route to specific hosts which fail to respond to the ping experiment?

Yes, it is possible. Firewalls often block ICMP Echo Requests (Ping) but allow UDP/TCP probes used by traceroute. Furthermore, traceroute relies on intermediate routers (which usually respond) rather than the final destination to map the path. Even if the final host is "silent," the path leading to it remains visible

d. Route Graphs

```

harshita@halcyon:~/Academics/CN/12340920 $ traceroute iitbhillai.ac.in
traceroute to iitbhillai.ac.in (192.168.10.115), 30 hops max, 60 byte packets
 1  _gateway (10.10.64.1)  8.065 ms  8.020 ms  7.982 ms
 2  10.200.10.14 (10.200.10.14)  7.734 ms  7.005 ms  7.625 ms
 3  192.168.10.115 (192.168.10.115)  7.583 ms  7.552 ms  7.528 ms
harshita@halcyon:~/Academics/CN/12340920 $ traceroute flipkart.com

```

```

harshita@halcyon:~/Academics/CN/12340920 $ traceroute disneyplus.com
traceroute to disneyplus.com (34.110.155.89), 30 hops max, 60 byte packets
 1  _gateway (10.10.64.1)  3.018 ms  4.420 ms  4.394 ms
 2  10.200.10.14 (10.200.10.14)  2.841 ms  2.807 ms  2.782 ms
 3  103.147.138.250 (103.147.138.250)  4.278 ms  4.239 ms  4.213 ms
 4  10.119.47.129 (10.119.47.129)  4.188 ms  4.161 ms  4.135 ms
 5  10.132.25.217 (10.132.25.217)  96.203 ms  96.179 ms  96.154 ms
 6  10.255.237.29 (10.255.237.29)  96.129 ms  89.855 ms  89.815 ms
 7  10.255.238.166 (10.255.238.166)  89.794 ms  89.776 ms  80.163 ms
 8  10.152.7.214 (10.152.7.214)  80.095 ms  80.058 ms  80.011 ms
 9  89.155.110.34.bc.googleusercontent.com (34.110.155.89)  79.872 ms  79.838 ms  72.14.204.62 (72.14.204.62)  79.787 ms

```

```

harshita@halcyon:~/Academics/CN/12340920 $
harshita@halcyon:~/Academics/CN/12340920 $ traceroute twitter.com
traceroute to twitter.com (162.159.140.229), 30 hops max, 60 byte packets
 1  _gateway (10.10.32.1)  2.312 ms  3.031 ms  2.868 ms
 2  10.200.10.14 (10.200.10.14)  3.143 ms  3.115 ms  3.094 ms
 3  103.147.138.250 (103.147.138.250)  3.431 ms  3.415 ms  3.399 ms
 4  static.ill.117.232.137.122.bsnl.co.in (117.232.137.122)  5.456 ms  5.440 ms  4.835 ms
 5  * * *
 6  * * *
 7  * * *
 8  117.216.207.105 (117.216.207.105)  95.741 ms  96.102 ms  95.881 ms
 9  162.158.226.28 (162.158.226.28)  95.783 ms  * *
10  162.158.226.21 (162.158.226.21)  95.721 ms  102.230 ms  162.158.226.71 (162.158.226.71)  102.110 ms
11  162.159.140.229 (162.159.140.229)  102.059 ms  101.963 ms  101.920 ms
harshita@halcyon:~/Academics/CN/12340920 $

```

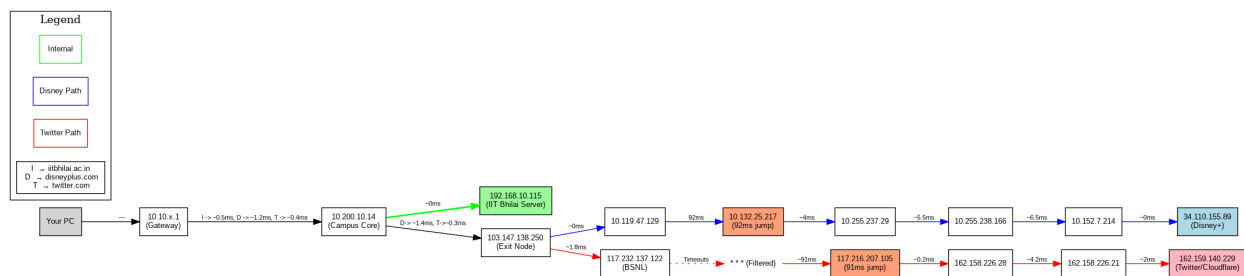


Figure 2: Graph

5 Question 5: NMAP

a. Usage

Nmap is a open source tool, which is used to discover devices on a network and gather information about them. It helps identify active hosts, open ports, running services, operating systems, and potential security vulnerabilities. Network administrators and security professionals use Nmap for network inventory, troubleshooting, and security auditing to understand how systems are exposed and to improve network security.

b. iitbhilai.ac.in Analysis

Port	Service	Usage
22/tcp	SSH	Used for secure remote login and server administration using encrypted communication.
80/tcp	HTTP	Used to provide web services and deliver website content over an unsecured connection.
443/tcp	HTTPS	Used to provide secure web services using SSL/TLS encryption.
5666/tcp	NRPE	Nagios Remote Plugin Executor, used for remote monitoring of system health and services.

```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx:~/Academics/CN/12340920
harshita@halcyon:~/Academics/CN/12340920 $ nmap iitbhilai.ac.in
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-09 22:49 IST
Nmap scan report for iitbhilai.ac.in (192.168.10.115)
Host is up (0.050s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
5666/tcp  open  nrpe

Nmap done: 1 IP address (1 host up) scanned in 10.17 seconds
harshita@halcyon:~/Academics/CN/12340920 $ nmap -sV iitbhilai.ac.in
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-09 22:49 IST
Nmap scan report for iitbhilai.ac.in (192.168.10.115)
Host is up (0.042s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 7.4 (protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.6 ((CentOS) OpenSSL/1.0.2k-fips PHP/5.4.16)
443/tcp   open  ssl/http     Apache httpd 2.4.6 ((CentOS) OpenSSL/1.0.2k-fips PHP/5.4.16)
5666/tcp  open  tcpwrapped

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.45 seconds
harshita@halcyon:~/Academics/CN/12340920 $
```

Version Detection: Yes, using `nmap -sV`. Nmap sends specially crafted packets to the open ports and analyzes the responses to match them against its service signature database. Based on this analysis, it determines the service type and its version, if available.

c. OS Detection

Observation: In this scan, Nmap attempted OS detection but reported “No exact OS matches for host”. However, the TCP/IP fingerprint information indicates that the host is running on a Linux-based operating system with an x86_64 architecture, as shown by the fingerprint field `P=x86_64-pc-linux-gnu`.

Process: Nmap performs OS detection by sending specially crafted packets to the target host and analyzing responses such as TCP sequence numbers, window sizes, and flag behavior. These characteristics are compared against Nmap’s OS fingerprint database to determine the closest matching operating system. If an exact match is not found, Nmap reports a fingerprint instead of a specific OS name.

```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx:~/Academics/CN/12340920
harshita@halcyon:~/Academics/CN/12340920 $ sudo nmap -O iitbhilai.ac.in

[sudo] password for harshita-patidar:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-09 23:24 IST
Nmap scan report for iitbhilai.ac.in (192.168.10.115)
Host is up (0.088s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   open  https
5666/tcp  open  nrpe
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/subm
t/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.94SVN%E=4%D=1/9%OT=22%CT=1%CU=34642%PV=Y%D5=3%DC=I%G=Y%TM=69614
OS:0D0%P=x86_64-pc-linux-gnu)SEQ(SP=103%GCD=1%ISR=10B%TI=Z%II=I%TS=A)SEQ(SP
OS:=103%GCD=1%ISR=10C%TI=Z%II=I%TS=A)SEQ(SP=FD%GCD=1%ISR=10F%TI=Z%TS=C)SEQ(
OS:SP=FD%GCD=2%ISR=10F%TI=Z%TS=C)SEQ(SP=FF%GCD=1%ISR=10E%TI=Z%II=I%TS=A)OPS
OS:(O1=M4E2ST11NW7%O2=M4E2ST11NW7%O3=M4E2NNT11NW7%O4=M4E2ST11NW7%O5=M4E2ST1
OS:1NW7%O6=M4E2ST11)WIN(W1=7120%W2=7120%W3=7120%W4=7120%W5=7120%W6=7120)ECN
OS:(R=Y%DF=Y%T=40%W=7210%O=M4E2NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=O%A=S+%F=A
OS:S%RD=0%Q=)T2(R=N)T3(R=N)T4(R=N)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=
OS:0%Q=)T6(R=N)T7(R=N)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RU
OS:CK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 3 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.82 seconds
```


d. Comparison with Other Sites

google.com: Observation:

- Open HTTP and HTTPS services found.
- The web services are running on Google Web Server (gws).
- Service versions and operating system details are could not be detected probaly because of security reasons.

```
harshita@halcyon: /Academics/CN/12340920 $ nmap google.com
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-10 12:13 IST
Nmap scan report for google.com (142.250.192.78)
Host is up (0.14s latency).
Other addresses for google.com (not scanned): 2404:6800:4009:829::200e
rDNS record for 142.250.192.78: bom12s16-in-f14.1e100.net
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp    open  https

Nmap done: 1 IP address (1 host up) scanned in 36.84 seconds
harshita@halcyon: /Academics/CN/12340920 $ nmap -sV google.com
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-10 12:13 IST
Nmap scan report for google.com (142.250.192.78)
Host is up (0.11s latency).
Other addresses for google.com (not scanned): 2404:6800:4009:829::200e
rDNS record for 142.250.192.78: bom12s16-in-f14.1e100.net
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      gws
443/tcp    open  ssl/https gws
2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF:Port80-TCP:V=7.94SVNMI=7X0=1/10%Tline=6961F546XP=x86_64-pc-linux-gnuNr(G
SF:etRequest,22C2,"HTTP/1.0\0x20200\0x200K\r\nDate:\x20Sat,\x2010\x20Jan\x2
SF:02026\0x2006:44:23\0x20GMT\r\nExpires:\x20-1\r\nCache-Control:\x20private
```

```
Nmap done: 1 IP address (1 host up) scanned in 112.46 seconds
harshita@halcyon:~/Academics/CN/12340920 $ sudo nmap -O google.com

[sudo] password for harshita-patidar:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-10 12:16 IST
Nmap scan report for google.com (142.250.192.78)
Host is up (0.061s latency).
Other addresses for google.com (not scanned): 2404:6800:4009:829::200e
rDNS record for 142.250.192.78: bom12s16-in-f14.1e100.net
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp    open  https
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.44 seconds
harshita@halcyon:~/Academics/CN/12340920 $
```

cloudflare.com: Observation:

- web and proxy-related services found.
- All detected services are Cloudflare HTTP proxy, confirming its role as a global CDN and security layer.
- Service versions and operating system details are could not be detected probaly because of security reasons.


```
harshita-patidar@harshita-patidar-HP-Pavilion-Laptop-15-eg3xxx:~/Academics/CN/12340920
harshita@halcyon:~/Academics/CN/12340920 $ nmap cloudflare.com

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-10 12:21 IST
Nmap scan report for cloudflare.com (104.16.132.229)
Host is up (0.081s latency).
Other addresses for cloudflare.com (not scanned): 104.16.133.229 2606:4700::6810:85e5 2606:4700::6810:84e5
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
8080/tcp  open  http-proxy
8443/tcp  open  https-alt

Nmap done: 1 IP address (1 host up) scanned in 21.53 seconds
harshita@halcyon:~/Academics/CN/12340920 $ nmap -sV cloudflare.com

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-10 12:21 IST
Nmap scan report for cloudflare.com (104.16.133.229)
Host is up (0.074s latency).
Other addresses for cloudflare.com (not scanned): 104.16.132.229 2606:4700::6810:85e5 2606:4700::6810:84e5
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
80/tcp    open  http      Cloudflare http proxy
443/tcp   open  ssl/http  Cloudflare http proxy
8080/tcp  open  http      Cloudflare http proxy
8443/tcp  open  ssl/http  Cloudflare http proxy

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.05 seconds
harshita@halcyon:~/Academics/CN/12340920 $ sudo nmap -O cloudflare.com

Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-10 12:22 IST
Nmap scan report for cloudflare.com (104.16.133.229)
Host is up (0.058s latency).
Other addresses for cloudflare.com (not scanned): 104.16.132.229 2606:4700::6810:84e5 2606:4700::6810:85e5
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
8080/tcp  open  http-proxy
8443/tcp  open  https-alt
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
OS fingerprint not ideal because: Missing a closed TCP port so results incomplete
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.45 seconds
harshita@halcyon:~/Academics/CN/12340920 $
```